Software Requirements Specification

SENG 271 TERM PROJECT — SRS V2

Project Title	Department Course Timetable Management System
Document Version	v2 (Revised Specification)
Date	10.11.2026
Team Number	Team 19
Team Members	Eren KARAKAŞ (@ErennKarakass) • Semih Mertcan İÇYER (@semihicyer)
Repository URL	https://github.com/ErennKarakass/Daviser.git

Revision History

Version	Date	Author(s)	Summary of Changes
v1	24.10.2026	Team 19	Initial requirements extracted from the project description.
v2	10.11.2026	Team 19	Revised requirements incorporating SDD v2 design decisions: refined file parsing validation rules, detailed conflict classifications, explicit hard/soft constraint specifications, and classroom request email draft requirements.
Final	[Pending]	Team 19	Final requirements verified against delivered application at Checkpoint 3.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for the Department Course Timetable Management System. It serves as the primary agreement between the development team and course evaluators, outlining system capabilities for automatic timetable generation, constraint enforcement, conflict handling, and manual schedule adjustments.

1.2 Scope

In Scope: Reading course, instructor, room, and constraint data from external files (CSV/TXT); representing four undergraduate year groups; automated weekly timetable construction using rule-based scheduling; detection and logging of scheduling conflicts; manual assignment modification and validation; separate reporting of room shortages; generation and disk export of formal classroom request email drafts.

Out of Scope: External database integration, web application frontends, network protocol communications, automatic background email sending via SMTP, student-level individual elective conflict handling, and automated solver libraries.

1.3 Definitions

Term	Definition
Session (Block)	A discrete theoretical (-T) or laboratory (-L) teaching unit belonging to a course, scheduled in consecutive one-hour time slots without splitting.
Time Slot	A standardized 1-hour scheduling unit spanning 09:00 to 17:00 on working days (Monday–Friday).
Year Group	One of the four undergraduate cohorts (First Year, Second Year, Third Year, Fourth Year). Mandatory courses within the same year group cannot overlap.
Hard Constraint	A mandatory operational rule (HC-01 to HC-09) that must never be violated in a valid timetable.
Soft Constraint	A secondary preference (SC-01 to SC-04) that should be satisfied when possible, triggering a warning if violated.
Special Request	A specific scheduling directive associated with a course (e.g., fixed time, designated laboratory room).
Unscheduled Session	A course session that could not be assigned to any valid slot and room due to constraint violations or room shortages.

1.4 References

- SENG 271 Term Project Assignment: Department Course Timetable Management System Document.
- SENG 271 Term Project Guide: Requirements, Design and Teamwork.
- SENG 271 Deliverable Templates & Repository Map.

2. Overall Description

2.1 Product Perspective

The system is a standalone Java console application. It operates by consuming plain text and CSV configuration files from the `data/` folder, building internal object models, generating weekly schedules via a custom rule-based scheduling algorithm, and rendering output timetables on console screens as well as exporting report files to the `output/` directory.

2.2 User Characteristics

User Role	Main Goals	Characteristics
Department Timetable Coordinator	Construct feasible timetables, resolve schedule conflicts, inspect year-group grids, manually adjust sessions, and export room request drafts.	

2.3 Operating Environment

The application executes within Java Runtime Environment (JRE) version 21 or 25 on standard desktop operating systems (Windows, macOS, Linux). It requires no external database or active internet connectivity.

2.4 Assumptions and Constraints

- Working days are Monday through Friday, operating strictly between 09:00 and 17:00 (8 one-hour slots per day).
- All sessions must finish by 17:00; overnight or weekend scheduling is prohibited.
- Each course belongs to exactly one primary year group (Years 1, 2, 3, or 4).
- Theoretical and laboratory hours for a course are treated as distinct blocks and scheduled consecutively without non-consecutive splits.

3. Functional Requirements

ID	Group	Requirement Description	Priority	Source	Screen
FR-01	Data Loading	The system shall load course curriculum data from `data/courses.csv`.	**Must**	§3.7	Screen 2
FR-02	Data Loading	The system shall skip invalid records in input files, log a detailed warning containing line numbers and values, and continue execution without crashing.	**Must**	§3.7	Screen 2
FR-03	Data Loading	The system shall load instructor details and availability periods from `data/instructors.txt`.	**Must**	§3.7	Screen 2
FR-04	Data Loading	The system shall load classroom and laboratory parameters from `data/rooms.csv`.	**Must**	§3.7	Screen 2
FR-05	Data Loading	The system shall load special scheduling requests from `data/constraints.txt` and classify them as mandatory or preferred.	**Must**	§3.6, §3.7	Screen 2

ID	Group	Requirement Description	Priority	Source	Screen
FR-06	Automatic Scheduling	The system shall automatically schedule sessions into valid time slots and rooms while satisfying all hard constraints (HC-01 to HC-09).	**Must**	§3.8	Screen 3
FR-07	Automatic Scheduling	The system shall prioritize processing fixed-time mandatory course requests before evaluating unconstrained sessions.	**Must**	§3.8	Screen 3
FR-08	Automatic Scheduling	The system shall attempt alternative valid time slots or rooms when a scheduling conflict is encountered.	**Must**	§3.8	Screen 3
FR-09	Conflict Handling	The system shall detect instructor time overlaps, room time overlaps, year-group session overlaps, and room capacity/type mismatches.	**Must**	§3.5, §3.8	Screen 3, 5
FR-10	Conflict Handling	The system shall mark sessions that cannot be validly placed as unscheduled and record an explicit failure reason.	**Must**	§3.8	Screen 5
FR-11	Timetable Viewing	The system shall display the weekly timetable for a selected year group in an 8-slot by 5-day grid layout.	**Must**	§3.9, §3.11	Screen 4
FR-12	Timetable Viewing	The system shall display detailed lists of all scheduled sessions across the department.	**Must**	§3.9	Screen 4
FR-13	Timetable Viewing	The system shall display all unscheduled sessions categorized by failure type (NO ROOM vs. CONSTRAINT).	**Must**	§3.9	Screen 5
FR-14	Timetable Viewing	The system shall display soft-constraint warning messages for accepted assignments that violate preferences.	**Should**	§3.5	Screen 3, 5
FR-15	Manual Modification	The system shall allow the coordinator to manually move an assigned session to a new day, time slot, and room.	**Must**	§3.9	Screen 6
FR-16	Manual Modification	The system shall re-evaluate all hard constraints during manual modifications and reject any change that introduces a hard constraint violation.	**Must**	§3.9	Screen 6
FR-17	Manual Modification	The system shall allow the coordinator to remove an existing assignment, returning the session to unscheduled status.	**Must**	§3.9	Screen 6

ID	Group	Requirement Description	Priority	Source	Screen
FR-18	Classroom Request	The system shall explicitly isolate unscheduled sessions caused solely by room shortages (NO ROOM).	**Must**	§3.10	Screen 5, 7
FR-19	Classroom Request	The system shall generate a formal classroom request email draft for a selected NO ROOM session detailing course code, student count, required room type, and suggested time.	**Must**	§3.10	Screen 7
FR-20	Classroom Request	The system shall allow the coordinator to save the generated email draft to a file in the `output/` directory.	**Must**	§3.10	Screen 7
FR-21	Output Generation	The system shall save generated weekly timetables and scheduling summary reports to text/CSV files in `output/`.	**Must**	§3.11	Screen 1

4. Scheduling Constraints

4.1 Hard Constraints (Mandatory Rules)

ID	Constraint	Rule Description	Required Data	Violation Message
HC-01	Instructor Conflict	An instructor cannot teach two sessions simultaneously.	Instructor ID, assigned time slots	`Instructor [ID] busy`
HC-02	Room Conflict	A room cannot host multiple sessions simultaneously.	Room ID, assigned time slots	`Room [ID] occupied`
HC-03	Year-Group Conflict	Mandatory courses belonging to the same year group cannot overlap.	Course year group, time slots	`Year [N] overlap`
HC-04	Working Hours	All sessions must occur within Mon–Fri, 09:00–17:00.	TimeSlot day & start/end hour	`Outside working hours`
HC-05	Room Capacity	Assigned room capacity must be ≥ session student count.	Course student count, Room capacity	`Insufficient capacity`
HC-06	Room Type	Room type must match session requirements (Classroom vs Laboratory).	Course room type, Room type	`Room type mismatch`

ID	Constraint	Rule Description	Required Data	Violation Message
HC-07	Instructor Availability	Sessions cannot be placed in instructor unavailable hours.	Instructor unavailable slots	`Instructor unavailable`
HC-08	Fixed-Time Requirements	Courses with predefined times must be assigned strictly to those times.	Constraints fixed day/time	`Fixed time conflict`
HC-09	Room Availability	Sessions cannot be placed during room maintenance/unavailable hours.	Room unavailable slots	`Room unavailable`

4.2 Soft Constraints (Preferences)

ID	Constraint	System Consideration Strategy	Warning Text
SC-01	Theory Before Laboratory	Attempts to place theory block (-T) on an earlier day or earlier slot than laboratory block (-L).	`SC-01: Lab scheduled before theory`
SC-02	Instructor Preferences	Attempts to place sessions in instructor preferred times (e.g. morning hours).	`SC-02: Instructor afternoon placement`
SC-03	Avoiding Gaps	Attempts to minimize idle gaps between year group sessions on the same day.	`SC-03: Gap detected for Year [N]`
SC-04	Consecutive Hours	Multi-hour blocks are scheduled in contiguous slots by default.	`SC-04: Non-consecutive slots tried`

4.3 Special Scheduling Requests

Request Type	Example Input	Kind	Handling Strategy
Fixed Time	SENG301 on Wed 13:00-16:00	Mandatory	Scheduled first in Step 1; reported as constraint failure if hard constraint fails.
Specific Room	SENG401 must use Lab B202	Mandatory	Limits room candidate selection strictly to the requested room ID.
Instructor Day Off	Dr. Demir unavailable Fri	Mandatory	Adds Friday slots to instructor unavailable list (enforced via HC-07).
Lab Before Theory	SENG302 Lab may precede Theory	Preferred	Suppresses SC-01 warning for the designated course.

5. Input Data Requirements

File Path	Format	Fields (in order)	Validation Rules
`data/courses.csv`	CSV	code, title, year, instructorId, theoryHours, labHours, students	Year must be 1-4; hours ≥ 0; students > 0; instructorId must exist in instructors.txt.
`data/instructors.txt`	TXT	id; name; unavailableSlots; preferredSlots	ID must be unique; slots formatted as Day HH:00-HH:00.
`data/rooms.csv`	CSV	id, type, capacity, unavailableSlots	Type must be 'Classroom' or 'Laboratory'; capacity > 0.
`data/constraints.txt`	TXT	courseCode; requestType; parameters; mandatoryFlag	Course code must exist in courses.csv; valid request types.

Input Error Handling

Error Condition	Example Scenario	System Reaction
Missing File	`rooms.csv` file not found in `data/`	Display readable error message and halt data loading gracefully without crashing.
Invalid Numeric Value	Theory hours specified as "three" in `courses.csv`	Log warning with filename, line number, invalid token; skip record and continue.
Incomplete Record	Course line missing students field	Log warning with line number; skip record.
Nonexistent Reference	Request referencing unknown course `SENG499`	Log warning; ignore invalid constraint request and continue loading.

6. Output Requirements

Output Item	Content Description	Console View	Saved File Location
Weekly Timetable (Year Group)	5-day x 8-slot grid showing course code, block type, room ID, and details list.	Yes (Screen 4)	`output/timetable_year[N].txt`
Scheduled Sessions List	Complete list of assigned sessions with instructor, day, time, and room.	Yes (Screen 4)	`output/scheduled_sessions.txt`

Output Item	Content Description	Console View	Saved File Location
Unscheduled Sessions Report	Sessions failing assignment with problem classification (NO ROOM vs. CONSTRAINT).	Yes (Screen 5)	`output/unscheduled_report.txt`
Classroom Request Draft	Formal email text to Student Affairs Office requesting additional room capacity.	Yes (Screen 7)	`output/classroom_request_[CODE].txt`

7. Non-Functional Requirements

ID	Category	Requirement Description
NFR-01	Reliability	The system shall not crash or terminate unexpectedly when encountering malformed input lines or invalid menu entries.
NFR-02	Usability	The system shall provide clear console menu options (Screen 1 to 7) and display descriptive error/warning messages.
NFR-03	Performance	The system shall execute data loading and timetable generation for sample datasets in under 3 seconds.
NFR-04	Maintainability	The Java code shall adhere to OOP principles, maintaining strict design-code consistency with SDD v2.

8. Changes from Previous Version (v1 to v2)

Section / ID	Change Type	Detailed Reason for Change
FR-02, FR-05	Refinement	Specified exact warning logging format (line number, filename, token) based on SDD v2 input handling design.
FR-18, FR-19	Added	Explicitly added classroom request email generation requirements (Screen 7) following coordinator feedback.
Section 4.1	Clarification	Added standard violation message strings for HC-01 through HC-09 to align error reporting across team code.
Section 5	Refinement	Added input validation error table to cover edge cases identified during Java skeleton development.

AI-Use Declaration

Tool	Used For	Verification Method
ChatGPT / Claude / Gemini	Assisting in document formatting, requirement table structuring, and grammar checking.	Both team members reviewed every sentence against the project assignment PDF to ensure 100% accuracy and compliance.